Vitruvian Recurrent Unit

Document 9

The Arithmetic Benchmark: Carry Propagation and the Scratchpad Hypothesis

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Abstract

Documents 1–8 established that the Vitruvian Recurrent Unit (VRU) outperforms Vanilla-RNN and matches LSTM on continuous sequence tasks through spectral insulation — phi = 4/pi geometrically decoupling the recurrent weight spectral radius from activation drift. Document 9 introduces a harder benchmark: multi-digit arithmetic. This task requires not just memory but self-referential computation — the output at each position depends on the output at the previous position (carry propagation). Two architectural extensions were developed (VRU v9 and v10), a four-probe inline diagnostic suite was built, and the fundamental limitation of continuous hidden state memory for discrete carry propagation was identified. The scratchpad hypothesis — emitting carry tokens as discrete working memory — is proposed as VRU v11 and represents the next experimental frontier.

1. Why Arithmetic

The copying memory task (Documents 3–5) was designed to test pure memory — hold a token, echo it back after a delay. VRU succeeded because phi's spectral insulation keeps the hidden state stable over long delays. The task has no computation dependency: knowing token t does not require knowing what was output at token t-1.

Multi-digit arithmetic is fundamentally different. Addition of two n-digit numbers requires carry propagation: whether digit position k produces a carry depends on the sum at position k, which depends on the carry from position k-1. This is a self-referential dependency chain — computation referencing its own previous output. The task directly tests whether VRU's geometric structure can sustain this class of computation.

Humans solve this with discrete working memory — explicitly holding the carry as a symbol ('carry the 1') and reading it at the next step. Current neural architectures must approximate this with continuous hidden states. Document 9 asks: can phi's geometric structure do what LSTM gates do for carry propagation?

2. Experimental Setup

2.1 Reversed Digit Format

Addition is presented in reversed digit order so that the least-significant digit appears first — matching the natural order of carry propagation. The number 921 becomes the token sequence '1', '2', '9'. This means the model encounters digits in the order it needs to process them, and must output carry-dependent digits in the same natural order.

Example encoding:

```
921 + 433 = 1354 → '129 + 334 = 4531'
```

2.2 Auto-Curriculum

Training proceeds through eight curriculum levels. Levels 1–3 and 5 and 7 cover addition/subtraction at 1–5 digit precision. Levels 4, 6, and 8 introduce multiplication at 2–4 digits. The model advances when answer-token accuracy exceeds 95% for at least 5 consecutive epochs.

Level	Operations	Digits	Purpose
L1	+, -	1	Baseline arithmetic
L2	+, -	2	Two-digit carry
L3	+, -	3	Three-digit carry chain
L4	*	2	Multiplication intro
L5	+, -	4	Four-digit carry chain
L6	*	3	Three-digit multiplication
L7	+, -	5	Five-digit carry chain
L8	*	4	Four-digit multiplication

Table 1. Eight-level arithmetic curriculum.

2.3 Answer-Only Accuracy Metric

Early training exposed a metric trap: measuring accuracy over all tokens (including the input prompt) gave artificially high scores because the model trivially predicts the input digits it just read. The metric was corrected to measure only tokens after the '=' sign — the actual answer. This is the ans_acc metric reported throughout.

2.4 Architecture

VRU v9 baseline: hidden_dim=512, 4 layers, ~2.1M parameters.

```
h_t = tanh( W_x * x_t + PHI * W_h * h_{t-1} )
```

Optimizer: Adam lr=1e-3. Scheduler: CosineAnnealingWarmRestarts (T_0=30) to prevent LR decay trapping the model in early plateaus. Gradient clip: 5.0. Spectral radius clipping via power iteration once per epoch.

3. Inline Probe Suite

Four diagnostic probes were embedded directly into the training loop, firing every 10 epochs and on every level transition. This gave real-time visibility into the model's internal state during training — no separate analysis pass required.

Probe	What It Measures	Key Question
[SPECTRAL]	phi*rho for W_h and W_c per layer	Is the field geometrically stable?
[MEMORY]	Recall accuracy at 1d–4d, with borrows	Where does memory break down?
[CARRY]	Accuracy vs carry count (c0–c3)	Is carry learned as a general op?
[GRADIENT]	Gradient norm per layer (W_h+W_c)	Vanishing or exploding?

Table 2. Four inline probes running during training.

The carry probe was particularly diagnostic: it generates addition problems stratified by required carry count (0 to 3 carries) and measures accuracy per stratum. If the model has genuinely learned carry propagation, accuracy should be roughly flat across strata. If it has memorized patterns, accuracy drops sharply with carry count.

4. Training Results

4.1 L1 and L2: Clean Mastery

Every training run mastered 1-digit and 2-digit arithmetic reliably. L1 (1-digit +/-) mastered by epoch 5–6 at >95% ans_acc. L2 (2-digit +/-) mastered by epoch 14–24 at >95% ans_acc. The spectral probe showed phi*rho near 1.0 throughout — the field was stable. The carry probe showed 0% at all carry counts (the model hadn't reached 3-digit problems yet, where meaningful carry probing begins).

4.2 L3: Consistent Plateau at 50–66%

Every run — v9 and v10, across multiple configurations — plateaued at 3-digit addition/subtraction between 50–66% ans_acc and failed to advance. The plateau was persistent across 70+ epochs with no upward trend.

Problem	Expected	Model Output	Error
921 + 433 = 1354	1354	1202	-152
632 + 377 = 1009	1009	1013	+4
779 + 794 = 1573	1573	1515	-58
857 + 192 = 1049	1049	1011	-38
585 + 118 = 703	703	759	+56
790 + 352 = 1142	1142	1166	+24

Table 3. Representative L3 errors. First 1-2 digits correct, then drift.

The pattern is consistent: first 1–2 digits of the answer are correct, then the model drifts. This is a carry memory failure — the model loses the carry signal by the time it generates the 3rd output digit.

4.3 Probe Findings at L3

Probe	Observation	Interpretation
[SPECTRAL]	phi*rho 1.2–1.7 for all layers	Field marginally unstable despite clipping
[CARRY]	c0:0% c1:0% c2:0% c3:0%	Carry not learned as general operation
[GRADIENT]	10x drop L0→L3, VANISHING flag	Signal not reaching early layers
[MEMORY]	Fails at 1d (probe format mismatch)	Probe format bug — fix in v11

Table 4. Inline probe findings during L3 training.

Key finding: the carry probe showing 0% at all carry counts throughout L3 training confirms the model is not learning carry as a general algorithm. It is learning positional patterns that break down when carry chains exceed 2 digits.

5. VRU v10 — Self-Referential Carry Register

The theoretical motivation: the Vitruvian Man's proportions are not pairwise — every proportion references the navel, the geometric center. In an RNN processing arithmetic, the carry IS the self-referential point: the output of each digit position references the output of the previous position. VRU v9 has no explicit channel for this dependency.

VRU v10 adds W_c — a third weight matrix for the carry channel:

```
h_t = tanh( W_x*x_t + PHI*W_h*h_{t-1} + PHI*W_c*c_{t-1} )
c_t = h_t (carry = self-referential point from previous depth)
```

Component	VRU v9	VRU v10
Recurrent weights	W_h only	W_h + W_c
Carry channel	Implicit in h	Explicit W_c * c_{t-1}
State tuple	(h)	(h, c)
Parameters	~2.1M	~3.1M (+48%)
W_h initialization	Orthogonal	Orthogonal
W_c initialization	N/A	Orthogonal
Spectral clipping	W_h only	W_h and W_c

Table 5. VRU v9 vs v10 architecture comparison.

v10 showed identical training dynamics to v9 — same plateau at L3, carry probe still 0% throughout. The carry register is architecturally correct but the implementation is flawed: c = h_t means the carry channel carries the full hidden state, which is too noisy. Input embeddings, layer norms, and residual connections all contaminate the carry signal.

> The W_c register did not improve L3 accuracy because c = h_t is not a clean carry signal. The carry information is diluted by every other component of the hidden state. The architecture is right; the signal being carried is wrong.

6. Root Cause Analysis

The L3 plateau is not a stability problem, a capacity problem, or a learning rate problem. All three were addressed (spectral clipping, 512 hidden dim, warm restarts) without effect. The root cause is architectural:

Multi-digit addition in reversed format requires the model to generate digit k based on the sum of input digit k plus the carry from digit k-1. In the reversed token sequence for a 3-digit problem, generating the 4th answer token requires information from the 1st input token — a span of 10+ positions. The carry from digit k-1 must be encoded somewhere in the hidden state for 3 timesteps while the model reads the remaining input digits.

This is not a long-range memory problem (VRU handles seq_len=15,000 on continuous tasks). It is a precision problem: the carry is one bit of information (0 or 1) that must survive undistorted through 3 timesteps of a 512-dimensional hidden state that is simultaneously processing unrelated input tokens. The phi attractor preserves the overall spectral structure but cannot selectively preserve a single bit of information against overwriting.

Architecture	Carry Mechanism	Why It Works / Doesn't
Transformer	Full attention	Directly attends to carry position. No memory needed.
LSTM	Cell state + input/forget gates	Learned gates protect carry cell from overwriting.
VRU v10	W_c (noisy, c=h)	Carry drowned by other hidden state content.
VRU v11 (prop.)	Scratchpad token	Carry emitted as discrete token, read back explicitly.

Table 6. Carry propagation mechanisms across architectures.

7. The Scratchpad Hypothesis — VRU v11

Humans carry the 1 as a discrete symbol. Not a continuous-valued memory trace — a symbol. It either happened or it didn't. The scratchpad approach makes this explicit in the token stream.

7.1 Proposed Token Format

Current format (implicit carry):

```
129 + 334 = 4 5 3 1
```

Scratchpad format (explicit carry tokens):

```
129 + 334 = 4 5 3 1
```

Where = no carry from this position, = carry. The model is trained to predict both the digit and the carry state as a pair. At the next position, it reads both back — the digit as context, the carry token as the explicit working memory input.

7.2 Relationship to VRU v10

The W_c register in v10 was the right idea but carried the wrong signal. With scratchpad tokens, W_c processes a clean 2-token vocabulary embedding (carry/no-carry) rather than the full 512-dimensional hidden state. The geometric self-reference is preserved at the architectural level; the signal being referenced is purified to its essential content.

7.3 Theoretical Grounding

The carry token is the Transformative Middle made explicit. In the DPPU framework, the Transformative Middle is the shared state between two interacting systems — the information that passes between them at their boundary. In arithmetic, the carry IS that boundary state: it is what position k hands to position k+1. Making it a discrete token doesn't abandon the geometric framework. It implements it faithfully.

Relationship to Documents 7–8: phi provides spectral insulation of the hidden state between carry events. The carry token provides the discrete anchor that phi's continuous geometry needs to reference. The two mechanisms are complementary.

7.4 Why This Distinguishes VRU From LSTM

LSTM uses three sigmoid gates to protect carry information from overwriting. The gates have no geometric meaning — they are learned filters with no theoretical basis. VRU v11's scratchpad uses a different strategy: don't protect the carry in continuous memory, emit it as a symbol and read it back. This is closer to how the computation actually works, requires no gates, and produces an observable, interpretable carry trace.

Metric	VRU v10 Current	VRU v11 Target
L3 ans accuracy	50–66%	>95%
Carry probe c0	0%	>90%
Carry probe c1	0%	>80%
Carry probe c2	0%	>70%
Carry probe c3	0%	>60%
Spectral phi*rho	1.2–1.7	<1.05
Memory probe 3d	Fails	Passes
Carry token visible	No	Yes — interpretable

Table 7. Success criteria for VRU v11 scratchpad implementation.

8. Connection to Prior Theory (Documents 7–8)

Documents 7 and 8 established the spectral insulation mechanism: phi decouples W_h spectral radius from activation drift, providing gradient stability on continuous sequence tasks. Document 9 reveals the boundary of that mechanism.

Spectral insulation is a continuous-domain mechanism. It prevents the hidden state from being distorted by noise during long sequences. But carry propagation is a discrete-domain operation — one bit per digit position. A continuous insulator cannot selectively protect a single bit against the overwriting pressure of processing subsequent tokens.

This is not a failure of the phi mechanism — it is a scoping finding. Phi handles continuous memory; discrete computation requires discrete memory. VRU v11 combines both: phi for the continuous hidden state, scratchpad tokens for the discrete carry events. This is the first architecture in the DPPU framework that explicitly handles both.

The geometric chain extends:

```
Da Vinci: phi = 4/pi (circle-square ratio)

Documents 1-8: phi insulates W_h spectral radius from activation
drift

Document 9: discrete carry events need discrete tokens, not
continuous memory

Document 10 (proposed): phi + scratchpad = full arithmetic
mastery
```

9. Summary

Document 9 introduces arithmetic as a benchmark for self-referential computation in VRU. Key contributions:

The four-probe inline diagnostic suite (spectral, memory, carry, gradient) provides real-time visibility into training dynamics and was essential for diagnosing the L3 plateau.

The consistent L3 plateau across all runs (50–66% ans_acc) is diagnosed as a discrete carry memory problem, not a stability, capacity, or optimization problem. The carry probe showing 0% at all carry counts throughout training confirms the model has not learned carry as a general algorithm.

VRU v10's W_c carry register is architecturally correct but carries a noisy signal (c = h_t). The improvement required is not more capacity but signal purity.

The scratchpad hypothesis proposes emitting carry tokens as discrete working memory. This does not require any change to the phi equation or VRU cell structure — it changes what the model predicts. The carry token becomes the self-referential point from the previous depth, made discrete and observable.

> The boundary of phi's spectral insulation mechanism has been found: it provides continuous memory stability but cannot substitute for discrete symbolic working memory. VRU v11 combines both. If the scratchpad approach succeeds, it establishes a third paradigm beyond

LSTM (gated continuous memory) and Transformers (global attention): geometric continuous memory + discrete symbolic carry events.

Document 9 of the Vitruvian Recurrent Unit series. Follows Documents 1 (Origin), 2 (Field Simulation), 3 (RNN Experiments), 4 (LSTM Comparison), 5 (Stress Test), 6 (Comprehensive Benchmark), 7 (Why It Works), 8 (Probe Journey). Next: Document 10 will record VRU v11 scratchpad results.